



Alex Pavel Medina Jorquera

DATA SCIENTIST · SOFTWARE ENGINEER

Santiago, Chile

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Summary

Data Scientist and former Service Reliability Engineer with a background in Mathematical and Computational Engineering and a Master's in Computer Science. Experienced in machine learning, backend development, and cloud infrastructure. Proven ability to build end-to-end data pipelines, NLP classifiers, and production microservices.

Experience

Zipedi

Santiago, Chile

SERVICE RELIABILITY ENGINEER

Apr 2023 – Apr 2026

- Maintained operational reliability across a fleet of ~200 autonomous retail robots running near-simultaneously across client sites.
- Reduced incident response time from days to hours by designing and deploying automated alert systems that surfaced hardware/software faults.
- Developed production microservices on cloud, including statistical threshold models for proactive anomaly detection.

CORFO — Corporación de Fomento de la Producción

Santiago, Chile

MACHINE LEARNING INTERN

Jan 2022 – Jul 2022

- Built a text-based attribute classifier for innovation projects using scikit-learn and TensorFlow.
- Expanded available labelled data through automated multi-dimensional classification, improving model training coverage.

Supreme Court of Chile / IMFD — Millennium Institute Foundational Research

Santiago, Chile

JUNIOR DEVELOPER

Jan 2021 – Mar 2021

- Developed a Flask REST API enabling selection, training, and inference of machine learning models for legal document analysis.

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JUNIOR DATA SCIENTIST

Aug 2020 – Dec 2020

- Designed similarity metrics to detect and flag duplicate innovation project proposals.
- Implemented text vectorisation pipelines in Python (Pandas, scikit-learn) to enrich features used in predictive models.

Education

Pontifical Catholic University of Chile

Santiago, Chile

M.Sc. IN ENGINEERING SCIENCES — COMPUTER SCIENCES

Mar 2022 – Jan 2025

Specialization: Machine learning

Pontifical Catholic University of Chile

Santiago, Chile

PROFESSIONAL DEGREE IN MATHEMATICAL AND COMPUTATIONAL ENGINEERING

Mar 2021 – Jan 2025

Specialization: Data Science, Minor: Uncertainty Quantification

Pontifical Catholic University of Chile

Santiago, Chile

B.Sc. IN ENGINEERING SCIENCES

Mar 2015 – Dec 2020

Skills

Programming	Python, SQL, JavaScript, HTML/CSS
ML / Data	TensorFlow, Scikit-learn, Pandas, NumPy, Seaborn
Backend & APIs	Flask, REST APIs
Cloud & DevOps	GCP, Docker, Git/GitHub
Languages	Spanish (Native), English (B2), Dutch (A1)

Research

Comparison of MILP, SAT and ASP for Minimal Automata Synthesis from Traces

MASTER'S THESIS

Aug 2021 – Jan 2025

- Evaluated and benchmarked three formal methods — Mixed Integer Linear Programming, Boolean Satisfiability, and Answer Set Programming — for synthesising minimal deterministic finite automata from labelled example traces.